Project Status Report

AI Universal Receptionist - what is done, what is pending, what we added

Measured against BRD v2.0 | branch feat/mock-fitness-domain | 15 October 2026

At a glance

Four views of the same project: what is **completed**, what is **pending** from the BRD, what we **added beyond the BRD**, and where the implementation **differs from the BRD** (mismatches). Overall, 3 of 26 BRD requirements are fully done and 12 are partly done; 11 have not been started.

15	11	7	8
Completed (3 full + 12 in part)	Pending (not started)	Additional (not in BRD)	Mismatches with BRD

1. Completed

Working in the code today

Area	What is done	BRD ref
Conversation engine	Domain-neutral engine: prompt built only from the domain profile, streaming LLM replies, tool-use loop with error handling; a test blocks hotel wording inside the engine	O1, FR-20
Grounded FAQ answers	Answers only from the knowledge base or tool results; fixed refusal sentence when unknown; tested	FR-03
Conversation memory	Last 40 messages plus a context object reloaded every turn; AI does not re-ask known details	FR-20
Voice (browser)	Mic to Deepgram live speech-to-text, LLM, sentence-by-sentence text-to-speech, barge-in (caller can interrupt), text chat fallback	FR-02 (part)
Multi-tenant isolation	Every tenant table has tenant_id; repositories cannot be built without a tenant; tenant comes from the login token, never from the request; cross-tenant ids return 404	FR-24, O11
Roles and login	superadmin, admin, customer roles; Argon2 passwords, 15-min JWT, rotating refresh cookie with replay revocation, login throttling, email OTP on first login	FR-25 (part)
Business provisioning	Superadmin creates a business plus its admin, emails the generated password, can disable or enable a business (locks out admin and customers)	FR-21, 26 (part)
Domain profiles	Profile model (identity, FAQs, info, intents, booking fields, enabled tools); 8 domain templates (hotel, school, hospital, restaurant, real estate, salon, fitness, automobile)	FR-01 (part)
Booking via tenant systems	AI checks availability, books, looks up and cancels through each tenant's MCP server, confirming details first; proven with HolidayBooking and GymBooking mocks	FR-04, 05 (part)
Conversation logging	Conversations and timestamped messages saved per tenant/user; history list and transcript view in the UI	FR-08, 17 (part)
Admin web app	React app: login/register, OTP, dashboard, MCP settings, conversations, chat/voice console, superadmin businesses and users pages	-

Area	What is done	BRD ref
Quality	About 96 backend tests plus frontend tests; fake LLM/STT/TTS for offline testing; Docker Compose setup	-

2. Pending

BRD requirements not started, or only partly done

A. Not started (11 requirements)

FR	Pending item	Why it matters
06, 11	Call transfer (warm/cold), escalation after 2 failed attempts, human fallback that really connects	Today the AI only says it will connect a person
07	SMS and WhatsApp confirmations, changes, reminders	No customer notification after booking
09	After-hours handling and operating hours	24/7 promise has no hours logic
12, 13	Outbound calls: internal availability check, reminders, callbacks, no-show follow-up	Key BRD differentiator
15	Waitlist with automatic call when a slot opens	Differentiator
16	Sentiment and emergency/urgency detection	Safety-relevant for hospital domain
18	Smart hold queue	Overflow handling
19	Post-call feedback	Quality signal
22	Subscription plans, limits, upgrade/downgrade (simulated)	No commercial model
23	Usage metering (minutes, bookings, profiles) and dashboard	Needed for plans and billing

B. Started but incomplete (12 requirements): remaining work

FR	Done	Still pending
01	Profile model and templates	Create/edit UI and API, live switch, escalation and notification sections, several profiles per tenant
02	Greeting, streaming voice	Real phone calls (Twilio), number-to-tenant mapping
04, 05	Book, check, cancel via MCP	Workflow state machine (empty folder), reschedule, calendar, platform double-booking guard
08	Messages stored	Populate intent and outcome, write tool_calls log, mask caller, export
10	Language setting	Auto-detection, second language, non-English voice
14	Per-user history	Recognise caller by phone, greet by name, inject last booking
17	Text transcripts	Audio recording, disclosure, per-profile toggle
21	Superadmin provisioning	Self-service sign-up, guided setup, phone number provisioning
25	Three roles	Owner / Admin / Operator, tenant user management
26	Create, list, disable businesses	Health view, impersonation with audit log, offboarding and purge

C. Housekeeping: update docs/ARCHITECTURE.md and README (still describe a single-hotel Phase 0); the in-memory login throttle is per process only; the knowledge base lives in the prompt and will need retrieval as it grows.

3. Additional work (done beyond the BRD)

Valuable features the BRD never asked for

Addition	What it gives us
Per-tenant MCP integration	Each admin connects up to 5 MCP servers (URL, transport, custom headers, timeout); the AI sees only that tenant's tools. Secrets Fernet-encrypted, SSRF guard, no redirects, a broken server never fails a call
Marketplace / compare-all assistant	One shared assistant per domain asks all partners in parallel (read-only calls), caches answers; booking or cancelling must name exactly one partner
Customer role + email OTP	End customers self-register, verify by OTP, pick a business, chat or talk, and see only their own history
Domain catalog (8 domains)	Master data that auto-creates a starter profile and booking fields when a business is added
Pluggable AI providers	LLM via Groq, Anthropic or any OpenAI-compatible endpoint; swap with config
Mock customer systems	HolidayBooking (hotels, multi-hotel users) and GymBooking (classes): own API, web UI, MCP server, migrations, tests
Security hardening	Argon2, rotating refresh tokens with replay detection, login throttling, secret encryption

4. Mismatches: BRD vs. what we built

Same goal, different approach. Each needs a decision: update the BRD or change the code

Topic	BRD says	We built	Suggested decision
Call channel	Inbound phone calls via Twilio to a provisioned number	Browser microphone over WebSocket; Twilio used nowhere (SendGrid email only)	Add Twilio adapter, or accept browser voice for the POC
Booking data	Platform DB + each tenant's Google Calendar	Tenant's own system through MCP tools; no Google Calendar anywhere	Keep MCP model and amend BRD, or add a calendar connector
Tenant sign-up	Self-service sign-up and plan selection	Superadmin creates the business and emails the admin a password	Amend BRD, or add a public sign-up flow
Roles	Super-Admin, Owner, Admin, Operator	superadmin, admin, customer (customer is not in the BRD)	Add Owner/Operator, add customer to BRD
Intent detection	NLU against a domain intent map	Intent list exists in profile but the LLM decides implicitly; intent never stored	Store intent per message, or drop from BRD
Domain profile	Includes escalation routes and notification templates	Only identity, KB, intents, booking fields, tools	Extend profile when transfer and notifications land
Notifications	SMS / WhatsApp in 60 s	Email only, for credentials and OTP	Add SMS provider
Product scope	Single receptionist per tenant	Adds marketplace comparison across partners	Add to BRD as a new objective, or park